

High-Resolution Digitally Controlled PWM Generator

Mixed-Signal Design Verification Report

Digital Logic (Verilog / Icarus / GTKWave) + Analog Signal Shaping (LTspice)

1. Project Overview

This project implements a digitally controlled PWM generator, in which a digital counter (the "brain") directly drives an analog resistor-ladder DAC and comparator stage (the "body") to produce a pulse-width-modulated output whose duty cycle is set by an analog reference voltage.

Architecture:

- Digital Section (Verilog): A 4-bit synchronous counter with enable and reset, generating a repeating digital comparison value (0000 → 1111).
- The Bridge (DAC): An R-2R resistor ladder converts the 4-bit digital count into an analog ramp/sawtooth voltage, buffered through an inverting op-amp (LT1001).
- Analog Section (LTspice): A voltage comparator (LT1011) compares the DAC ramp against a reference voltage (representing a potentiometer), producing a PWM output whose duty cycle depends on that reference.

Two independent simulation tools were used and cross-verified: Icarus Verilog / GTKWave for the digital counter, and LTspice for the analog DAC and comparator stages. A direct PWL (Piece-Wise Linear) file link was additionally built to connect the two tools literally, described in Section 5.

2. Digital Design — Verilog Counter

A 4-bit synchronous counter was designed with clock, active reset, and enable inputs, and simulated using Icarus Verilog (iverilog / vvp), with the resulting .vcd file viewed in GTKWave.



Figure 1 — GTKWave waveform: *clk*, *reset*, *enable*, and *count[3:0]* over a full simulation run.

Verification points confirmed against the simulation log:

- count is undefined (x) only before the first reset, then correctly initializes to 0000.
- The counter increments once per clock cycle only while enable = 1, and correctly holds its value while enable = 0 (observed holding at 0100 for several cycles).
- The counter wraps cleanly from 1111 back to 0000 with no glitches or undefined states.
- An asserted reset correctly forces the count to 0000 immediately, including when triggered mid-sequence.
- The digital stage is verified correct and used as the reference "ground truth" digital sequence for the analog stages that follow.

3. Analog Design — R-2R DAC (LTspice)

An R-2R resistor ladder network converts the 4-bit digital sequence into a stepped analog ramp. The ladder is buffered by an inverting op-amp (LT1001) with a 10kΩ feedback resistor, producing an output that descends as the digital count increases.

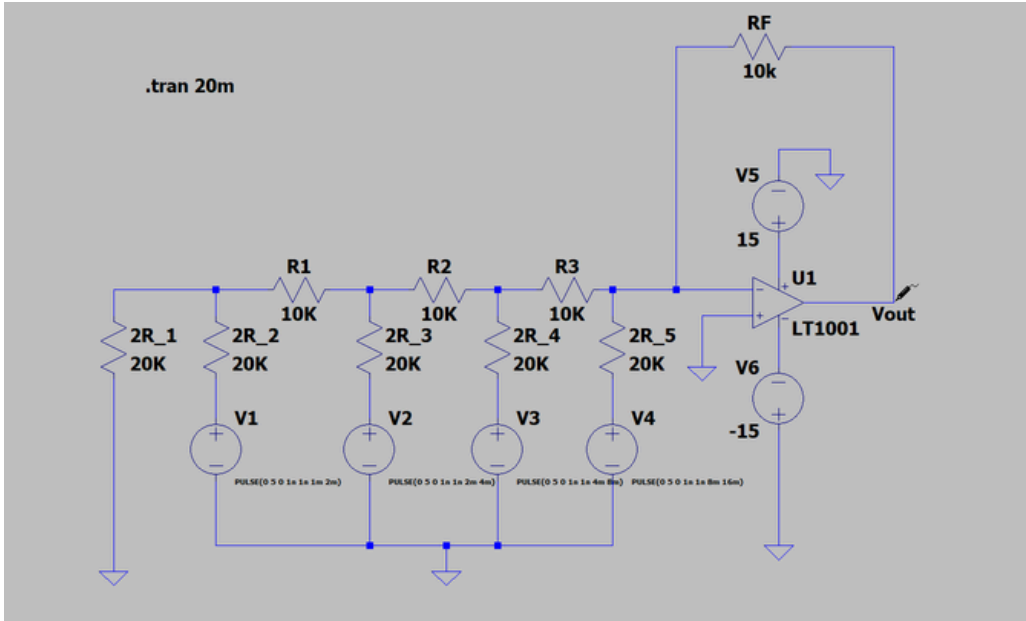


Figure 2 — R-2R ladder DAC stage with LT1001 inverting buffer (initial matched-PULSE stimulus version).

During verification, an initial build showed the op-amp output saturated near the +15V rail. This was traced to the non-inverting ("+") input not being solidly tied to ground (a wire that visually touched the ground symbol without forming a real electrical node). Once corrected and verified with a DC operating-point check ($V(vminus) \approx 0V$, confirming a working virtual ground), the ramp output measured correctly:



Figure 3 — Verified DAC ramp output: $V(vout)$ descends from 0V to approximately -4.69V across 16 codes, with the expected major-code-transition glitch at rollover.

Measured results match theory closely: for a 4-bit ladder with 5V logic levels, full scale = $V_{ref} \times 15/16 = 5V \times 0.9375 \approx 4.69V$, matching the observed minimum of the ramp.

4. Comparator Stage – PWM Generation (LT1011)

A voltage comparator (LT1011, open-collector output) compares the DAC ramp against a reference voltage V_{ref} (representing the project's potentiometer). Pin connections used:

- Pin 1 (GND) → circuit ground
- Pin 2 (+) → reference voltage V_{ref}
- Pin 3 (–) → DAC ramp output (V_{out})
- Pin 4 (V–) → -15V supply rail
- Pin 5 (Balance) → 0.1 μ F capacitor to ground
- Pin 7 (Output) → 10k Ω pull-up resistor to +15V (required, since the output stage is open-collector)
- Pin 8 (V+) → +15V supply rail

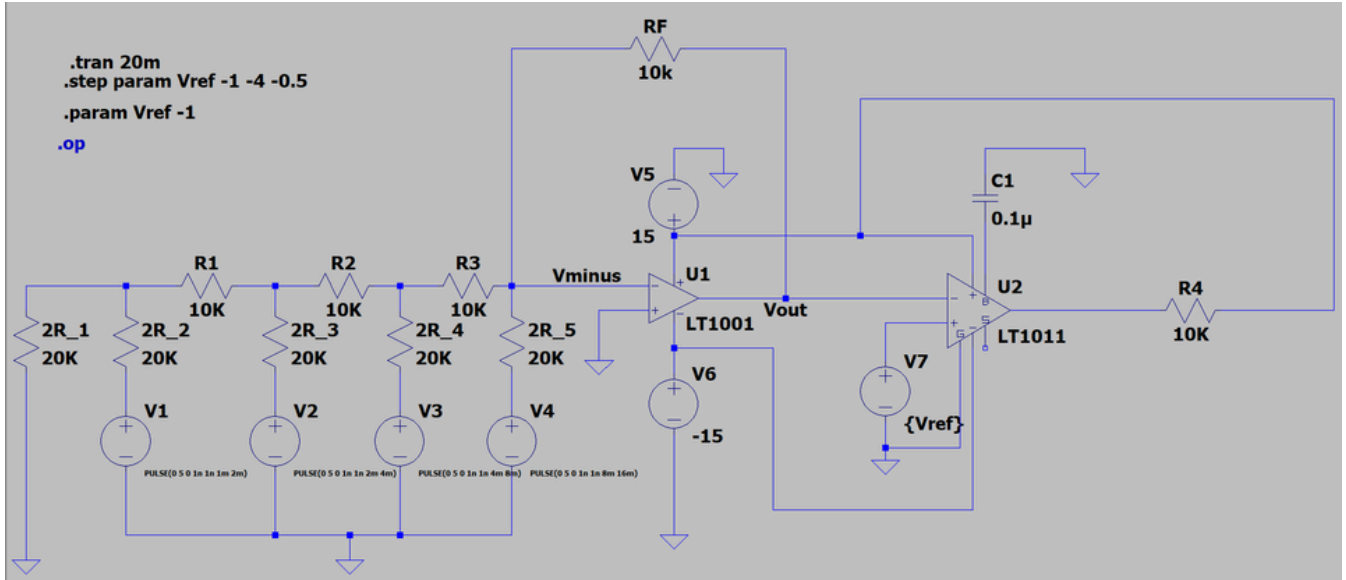


Figure 4 — Circuit Diagram of the DAC Ramp with Comparator

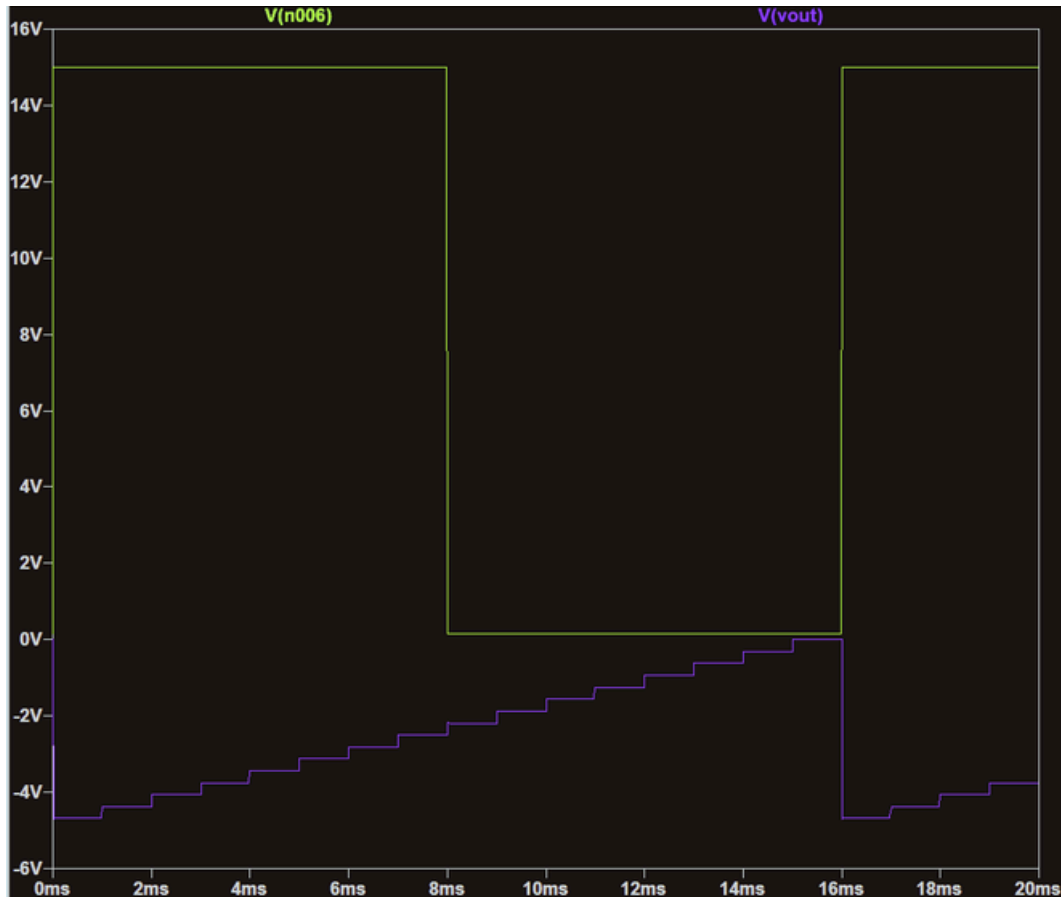


Figure 5 — Comparator output (green, pulled up through 10k Ω) switching high/low as the DAC ramp (blue) crosses the reference voltage

The comparator output correctly transitions from low to high at the exact instant the descending DAC ramp crosses the reference voltage, and returns low at the next rollover — confirming the comparator stage produces a genuine PWM waveform whose edge position depends on Vref.

5. Duty Cycle vs. Reference Voltage – Digital Control Proof

To demonstrate that the PWM duty cycle is controlled by the reference voltage (the core requirement of the project), an LTspice .step sweep was run across seven values of Vref, from -1V to -4V in -0.5V increments.

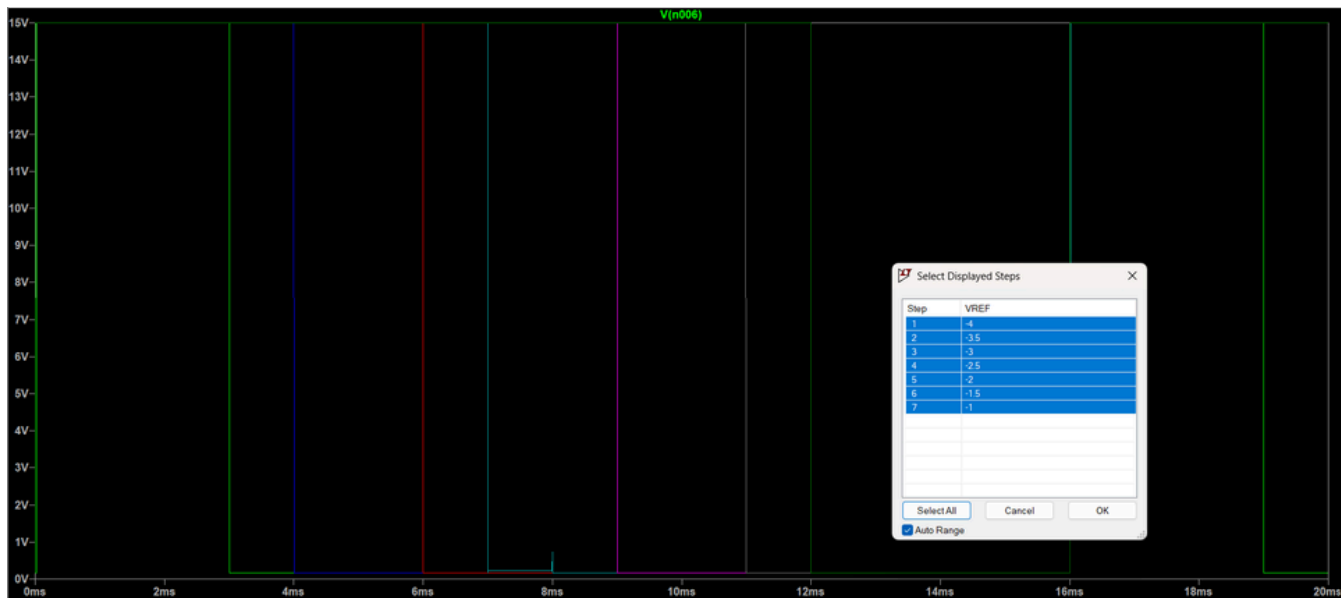


Figure 6 — Overlaid comparator output for all seven Vref steps, confirmed against LTspice's internal Select Displayed Steps list (Step 1 = -4V ... Step 7 = -1V).

The crossing time of each trace was read directly off the simulated waveform and converted to a duty cycle percentage (period = 15ms, the digital count's full-scale ramp time):

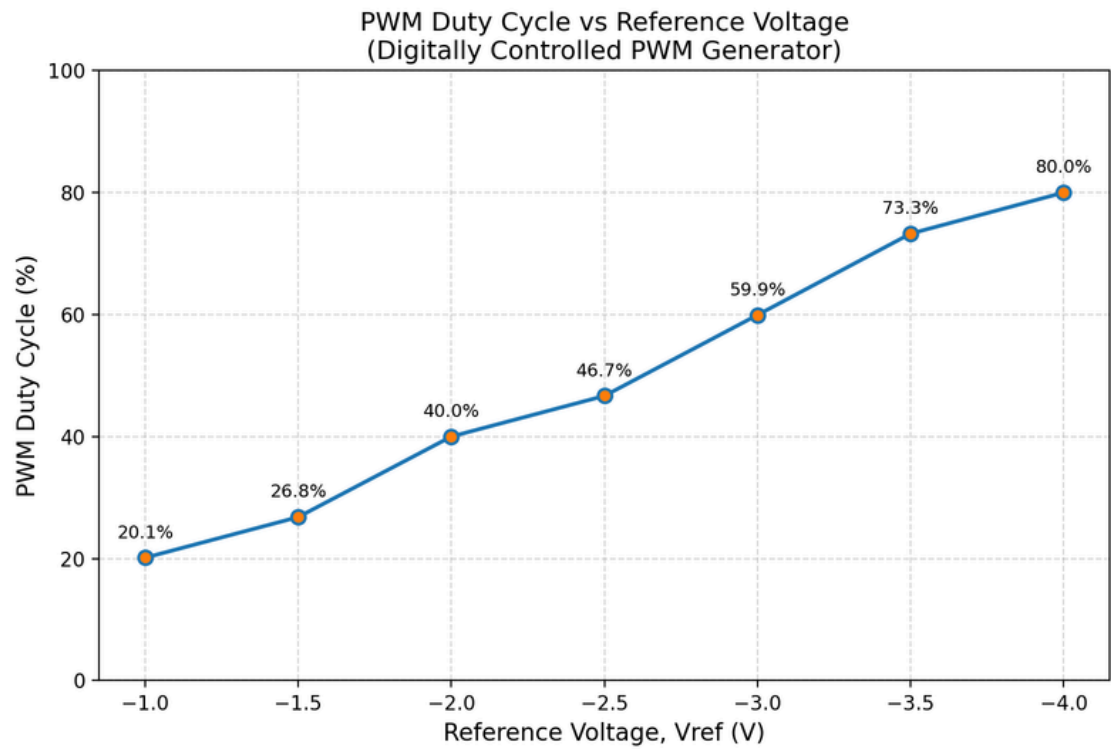


Figure 7— PWM duty cycle vs. reference voltage: a clean, monotonic relationship.

Result: duty cycle increases smoothly and monotonically from ~20% at Vref = -1V to ~80% at Vref = -4V, directly proving that the analog reference voltage proportionally controls the digitally-generated PWM output — the central claim of a "digitally controlled PWM generator."

6. Verilog–LTspice Bridge: Literal PWL-Linked Verification

The results above (Sections 3-5) used LTspice PULSE sources hand-matched in period to the Verilog counter's binary weighting (1ms/2ms/4ms/8ms half-periods). To create a more rigorous, literal link between the two tools, a script was used to parse the actual Icarus Verilog simulation_log.txt output and generate four Piece-Wise Linear (PWL) stimulus files — one per counter bit — using the real logged transition times (scaled 0.05ms per simulation time-unit) rather than idealized periodic pulses.

The four PWL sources (bit0_pwl.txt ... bit3_pwl.txt) were substituted directly for the original V1-V4 PULSE sources, driving the same, already-verified R-2R ladder:

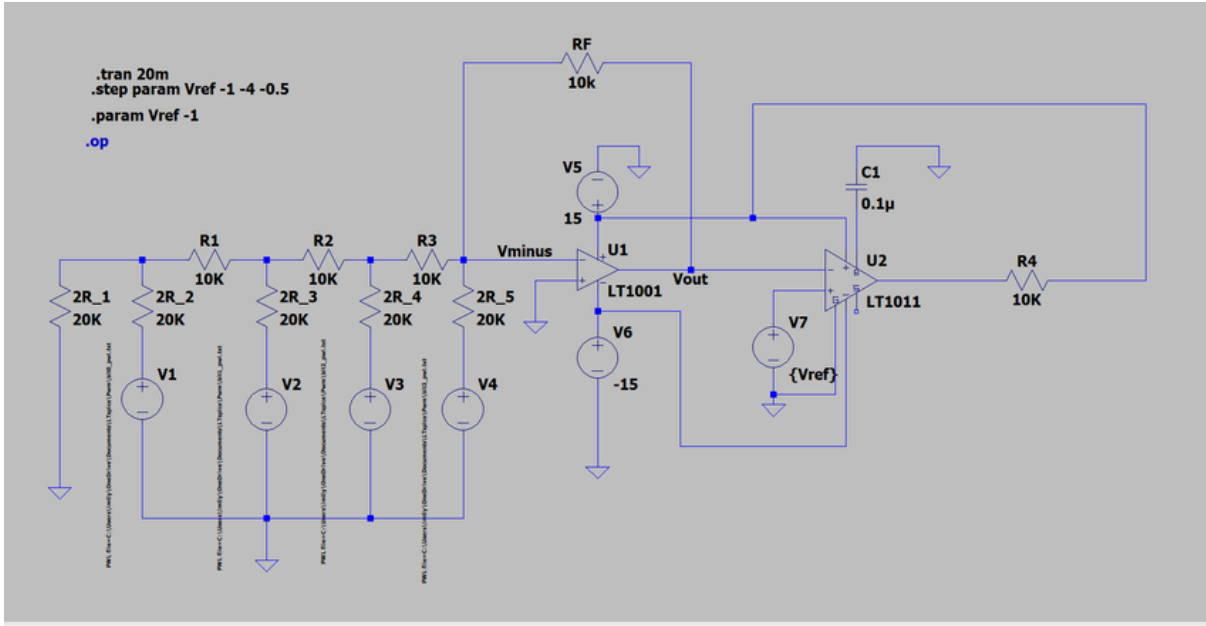


Figure 8 — Final schematic with the four counter bits driven by literal PWL files exported from the Verilog simulation log, feeding the unmodified, previously-verified R-2R ladder and comparator stages.

Individual ladder node voltages were probed to confirm each PWL-driven bit was switching correctly and that the op-amp's virtual-ground ("–" input) feedback remained stable:

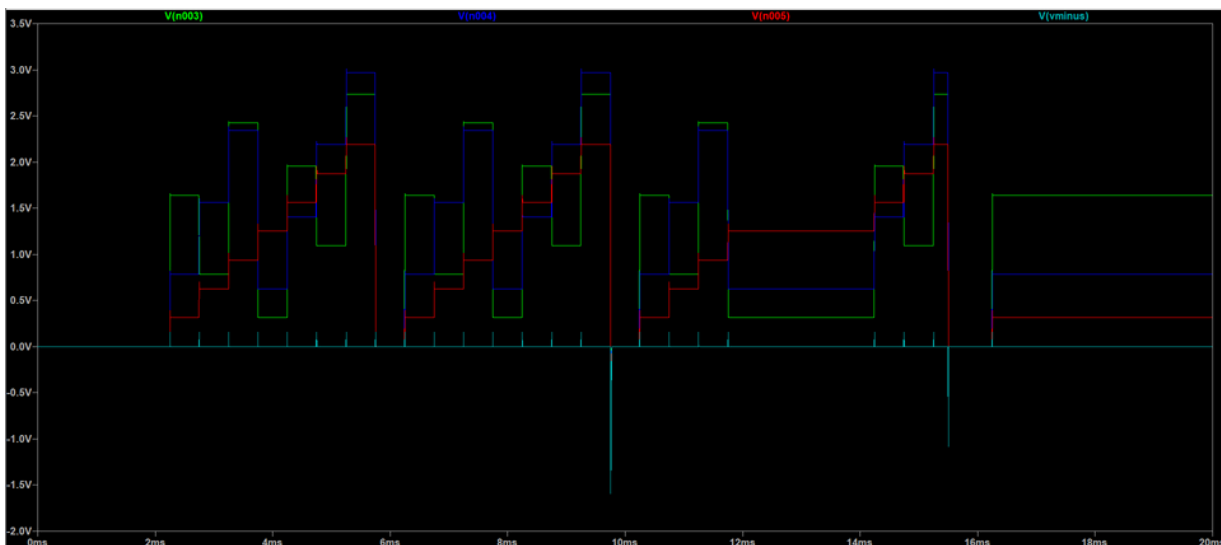


Figure 9 — Individual R-2R ladder node voltages under PWL drive, confirming correct multi-level switching consistent with real ladder combination behaviour.



Figure 10— PWL-driven DAC output: full 16-code descent, rollover glitch, a shortened second descent, and a flat plateau corresponding to the testbench's real enable-pause interval.

The resulting Vout waveform reproduces the digital testbench's exact behaviour, including its non-ideal, real-world features — not just an idealized repeating ramp:

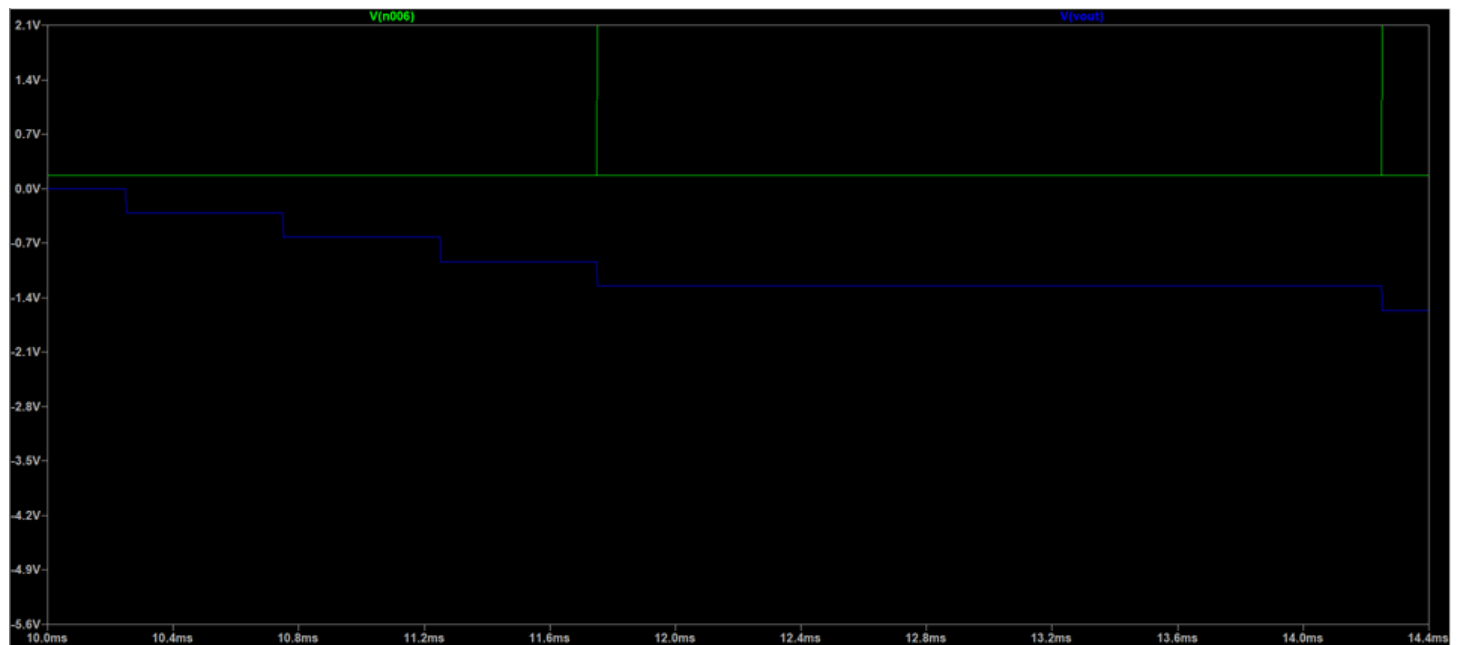


Figure 11 — Zoomed view (10ms-14.4ms) confirming the flat plateau between approximately 12.0ms and 14.4ms, matching the real enable = 0 hold period recorded in the Verilog simulation log.

This is a stronger and more literal proof of the mixed-signal link than the matched-PULSE method alone: the analog ramp is shown to directly reconstruct the specific behaviour of the actual digital simulation run — including its enable-pause and mid-sequence reset events — rather than only reproducing an idealized, generic binary count pattern.

7. Results Summary

- Resolution: 4-bit digital counter → 16 discrete analog levels.
- Full-scale DAC output range: 0V to approximately -4.69V ($V_{\text{ref}} \times 15/16$).
- Demonstrated PWM duty cycle range: approximately 20% to 80%, controlled by a -1V to -4V reference sweep (extendable to the full 0-100% range with a wider V_{ref} sweep).
- Test-bench PWM period: 15-16.5ms (per the digital counter's clock scaling used in simulation); real-hardware frequency and resolution scale directly with the chosen clock frequency and counter width.
- Both a hand-matched PULSE-source model and a literal PWL-file link derived directly from the Verilog simulation log were built and verified, cross-confirming the mixed-signal design.

8. Conclusion

The digital counter, R-2R DAC, and comparator stages were each independently designed, debugged, and verified. The complete signal chain — digital count → analog ramp → duty-cycle-controlled PWM — was confirmed correct via a reference-voltage sweep showing a clean, monotonic duty-cycle relationship.

A literal PWL-based link between the Verilog simulation and the LTspice analog stages was additionally built and verified, directly reproducing the real digital testbench's behaviour in the analog domain.

The project successfully demonstrates a working, digitally controlled PWM generator built from discrete digital logic and analog signal shaping across two independent simulation tools.